(PROJECT PROPSAL)

SMART INTERSECTION TRAFFIC LIGHT

THANADON TRAIPAK

KONGPHOP MEKAROON

NATTAWUT KAYACHAT

NORATHEP JAIKAEW

BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2022

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A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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IN COMPUTER ENGINEERING

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TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
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2022

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Abstract

Traffic congestion and accidents are significant problems in urban areas worldwide. The AI-powered smart traffic signal and pedestrian crossing system aim to address these issues by using computer vision and machine learning to make traffic and pedestrian flow management more efficient and adaptive. The smart traffic signal system detects the number of vehicles on each side of the road and adjusts the green light duration accordingly. If there are no vehicles on that side, the light will not turn green. and go to the green side with vehicles instead for the flow of traffic. The AI-powered pedestrian crossing system detects the number of pedestrians and vehicles at pedestrian crossings and adjusts the red light duration accordingly. It also alerts pedestrians when the signal is about to turn red, providing them with a warning to stop and let vehicles pass through. This system has the potential to significantly improve traffic flow and pedestrian safety in urban areas. It leverages the latest advances in computer vision and machine learning to create a more efficient and adaptive traffic management system that can benefit both drivers and pedestrians.

Keyword: Camera,Detects,Traffic Intersection Flow,Vehicle

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CHAPTER 1
INTRODUCTION

1.1 Background and Rationale

Traffic congestion and accidents are major problems in urban areas around the world. As cities grow and populations increase, traffic flow management and pedestrian safety become even more critical issues. Traditional traffic signals and pedestrian crossings are often static and do not adapt to changing traffic and pedestrian conditions. As a result, they can cause traffic congestion, delays, and accidents, which can have a significant impact on the economy and quality of life for people living in urban areas.

Automatic traffic lights that use cameras to turn red lights are one such solution. The rationale behind this technology is to use cameras to monitor traffic c at intersections and adjust the timing of traffic lights accordingly. This can help to reduce congestion and improve safety by optimizing the flow of traffic and reducing the likelihood of collisions.

However, there are potential risks and concerns associated with the use of cameras in traffic management systems, such as privacy violations. cyber attack and technical malfunctions It is therefore important to carefully assess the potential benefits and risks of using this technology. and to ensure that appropriate safeguards are in place to protect the privacy and security of individuals.

Overall, the AI-powered smart traffic signal and pedestrian crossing system have the potential to significantly improve traffic flow and pedestrian safety in urban areas. It leverages the latest advances in computer vision and machine learning to create a more efficient and adaptive traffic management system that can benefit both drivers and pedestrians.

1.2 Objective

1.2.1 To design and implement SMART INTERSECTION TRAFFIC LIGHT SYSTEM
1.2.2 To use the artificial intelligent software to simulate traffic condition for system testing
1.2.3 Develop the web application for controlling and supervising traffic lights.

1.3 Scope

1.3.1 Feature

1.3.1.1 The system can change the time of traffic lights in real time according to the number of vehicles.

1.3.1.2 Can detect vehicle by object detection technique (Computer vision)

1.3.1.3 This reduces the time spent waiting at traffic lights.

1.3.1.4 We use the library of the YOLO V8 to track vehicles and use supervision to get the ID of the track and another thing we use is prediction, and it together with supervision. When there is something that has a frame, it will count the vehicles.

1.3.1.5 We have used python. To create a camera and open it in the program

1.3.2 Develop a web application for scheduling traffic lights.
(a) The website will allow the admin to observe the traffic lights and cameras in real time.
(b) Dashboard: Create a dashboard that displays the status of traffic lights and provides controls to manage them.
(c) User authentication

1.3.3 Simulation

1.3.3.1 The traffic intersections created are based on the design and simulation of traffic intersections and are not based on actual locations (if you want to use them in a traffic intersection system, you need to set them up according to each intersection to get results. the best)

1.3.3.2 Simulate a traffic intersection with a total of 4 intersections and the number of lanes in each intersection equal to 4.

1.3.3.3 Simulate vehicle detection using camera images to prove detection capabilities.

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 System analysis

1.4.4 System design

1.4.5 Senior project 1 document preparation

1.4.6 Senior Project 1 exam

1.4.7 System development

1.4.8 System testing

1.4.9 Senior Project 2 exam

1.4.10 Senior Project 2 exam

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										
Senior Project 2 exam										

1.6 Expected Result

1.6.1 Develop a system that effectively manages traffic light time.
1.6.2 Have a test of the system's Traffic light intersection model.
1.6.3 Develop the web application to monitor traffic lights.
1.6.4 Get knowledge and implementation of the yolo object detection algorithm.

1.7 Place and Equipment

1.7.1 Place

1) Use a model of the traffic light intersection.
2) Simulation

1.7.2 Software

1) Python
2) Next.js
3) Yolo v8
4) Visual studio code
5) Obs studio

1.7.3 Hardware

1) Camera
2) Computer x 4

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

2.1.1 Neural networks

The basic idea behind neural network object detection is to train a model using large amounts of labeled data, which allows the network to learn to recognize the features that distinguish different objects. Once trained, the network can be used to detect objects in new images or videos by applying a sliding window approach or by using more sophisticated techniques such as anchor boxes.

2.1.2 Yolo

YOLO (You Only Look Once) is a real-time object detection algorithm that uses a single convolutional neural network to predict bounding boxes and class probabilities for objects in an image. The YOLO algorithm is fast, accurate, and has been widely used in various applications.

2.1.3 Other

One of the key components of a smart intersection traffic light and crosswalk light system is the traffic signal controller. This device controls the timing and sequencing of traffic signals, taking into account various factors such as traffic volume, pedestrian activity, and road conditions. By dynamically adjusting signal timing, the system can reduce traffic congestion, improve traffic flow, and enhance safety at intersections.

Another important component of the system is the pedestrian detection system. This technology uses cameras or sensors to detect the presence of pedestrians and cyclists at crosswalks. This information is then used to adjust the timing of traffic signals to provide more time for pedestrians to cross the street safely.

In addition to these components, smart intersection traffic light and crosswalk light systems may also include other features such as:

Vehicle detection and classification: sensors or cameras that can detect and classify vehicles, including cars, trucks, and bicycles, to optimize traffic flow and reduce congestion.

Intelligent traffic management: machine learning algorithms that can predict traffic patterns and adjust signal timing accordingly to minimize delays and improve efficiency.

Real-time information display: electronic signs or displays that provide real-time information to drivers and pedestrians, such as estimated wait times or traffic conditions ahead.

2.2 Related Research

"Improving Traffic Operations for Emission Reductions" by Koorosh Olyai and Sumit Sharma, published in the Transportation Research Record: Journal of the Transportation Research Board, discusses the relationship between traffic operations and carbon dioxide emissions and presents several strategies for reducing emissions through traffic management. The article also highlights the potential benefits of Intelligent Transportation Systems (ITS) in reducing congestion and improving traffic operations,leading to lower emissions. (Matthew Barth barth and Kanok Boriboonsomsin)[1]

The main objective of the Intelligent Traffic Light Control System based on Traffic Environment using Deep Learning is to develop a more efficient and flexible traffic light management system. The current traffic light management systems suffer from problems as they execute certain previously defined lines of code that cannot offer the adjustability of updation being executed in real-life scenarios. The proposed system aims to address these issues by using machine learning and deep learning technologies to make traffic lights more adaptive and responsive to real-time traffic conditions.(Moolchand Sharma1 ,Ananya Bansal, Vaibhav Kashyap, Pramod Goyal, Dr.Tariq Hussain Sheikh) [2]

The benefits of using a real-time traffic flow algorithm over traditional traffic lights are that it can efficiently schedule the increasing number of vehicles waiting to be processed at the intersection. The traffic signal phases are optimized according to collected data, mainly queue density and waiting time per vehicle, to enable as many as more vehicles to pass safely with minimum waiting time. This results in reducing congestion, less CO_2 emissions, and facilitates the movement of cars in intersections. [3]

2.3 Related Work

There are many studies for detecting road incidents for different environments, for example, freeway, highway, urbane, and non-urbane. The researchers contribute a lot to facilitate the community and focus on to minimize injuries and losses of human lives during driving on roads by designing efficient AID systems.(Zafar Iqbal and Majid Iqbal Khan 2018)[1]

These days we have cameras and wireless capability of detecting how many cars are passing and when they're passing. Once we have gathered this information live from the traffic stream the problem just becomes a big math equation where we're just looking for optimal green-red distribution and like any optimization problem the answer is intuitively simple, let artificial intelligence take over.
(S. S. ZIA++, M. NASEEM, I. MALA*, M. TAHIR, T. J. A. MUGHAL**, T. MUBEEN 2018)[2]

2.4 Related Technology

2.4.1 Software

2.4.1.1 Python

Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically typed and garbage-collected. We have used python. To create a camera and open it in the program

2.4.1.2 YOLO V8

Volo V8 is the highest performing YOLO model ever released, setting new standards in SOTA real-time detection and segmentation, opening up a new world of use cases for simple and effective AI solutions.We bring the library of yolo v8 to track vehicles and use supervision to get the ID of the track and another thing is prediction, bring it together with supervision. When there is something that has a frame, it will count the vehicles.

2.4.1.3 Visual Studio Code

Visual Studio Code, also commonly referred to as VS Code, is a source-code editor made by Microsoft with the Electron Framework, for Windows, Linux and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git.

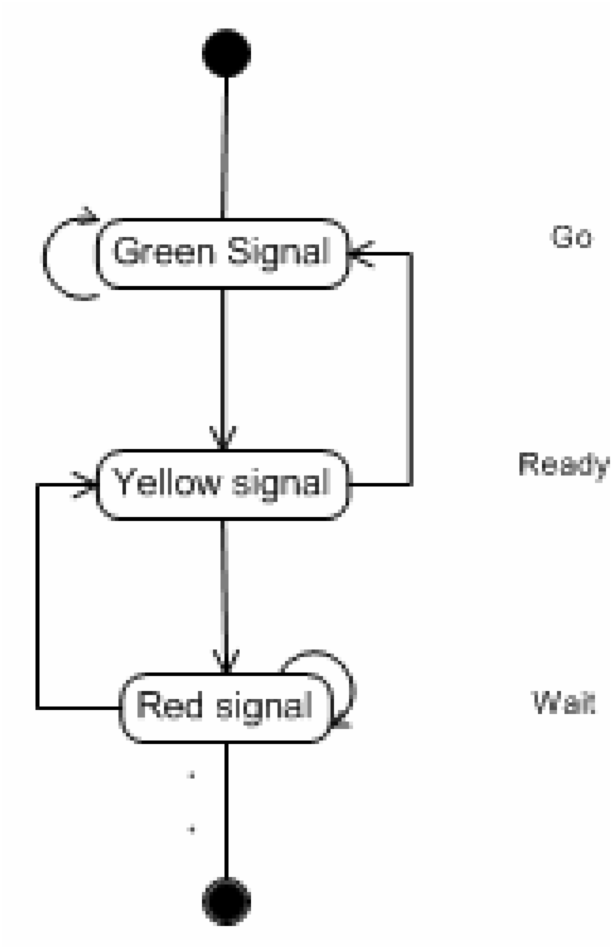
2.4.1.4 Next.js

Next.js is an open-source web development framework created by the private company Vercel providing React-based web applications with server-side rendering and static website generation.

CHAPTER 3

METHODOLOGY

3.1 Existing System



The existing could be demonstrated by a use case diagram in Figure 3.1.

Figure 3.1 Use case diagram of the existing system

In normal system will loop for endless and sometime system can be error and need to config on human

3.2 New System

In the intelligent traffic system, cameras are installed to count the number of vehicles caught at red lights and calculate the time for the vehicles to pass through the red lights in each round, resulting in more traffic flow because of the system. Slowly release adjacent cars to avoid the feeling of a traffic jam.

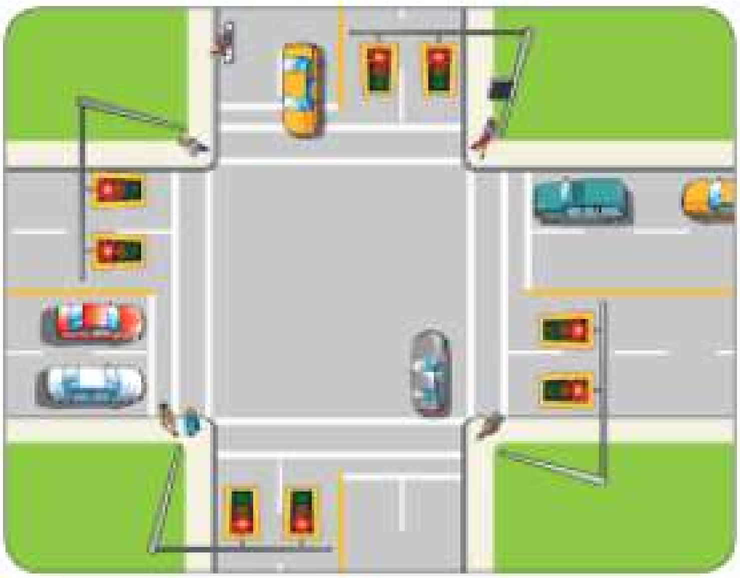


Figure 3.2.3 New system but use cameras to detect cars.

<u>analysis</u>

In the time from the research, the first row of cars will start taking time to exit the red light in 5 seconds and after that the car will take an average of 1.5 seconds per car per lane. According to the number of cars that have been stuck at the red light properly.

From the picture, there will be counting loops from intersections, for example from 4 intersections in front of simulation. When the green light is released in one lane, there will be time to count the number of vehicles in the lane waiting for the red light in order to calculate the time to prepare the release time for the next step

design from camera

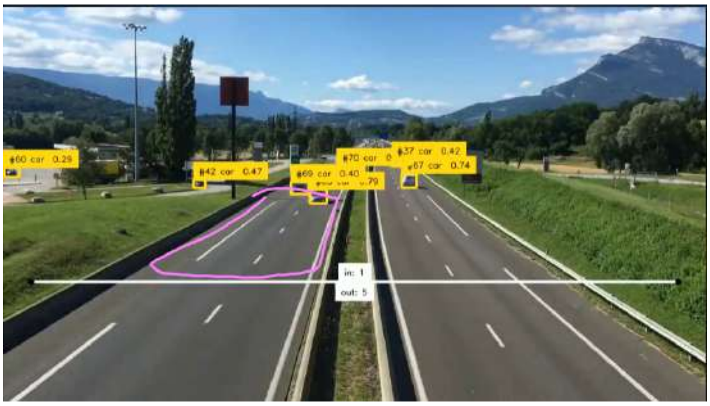
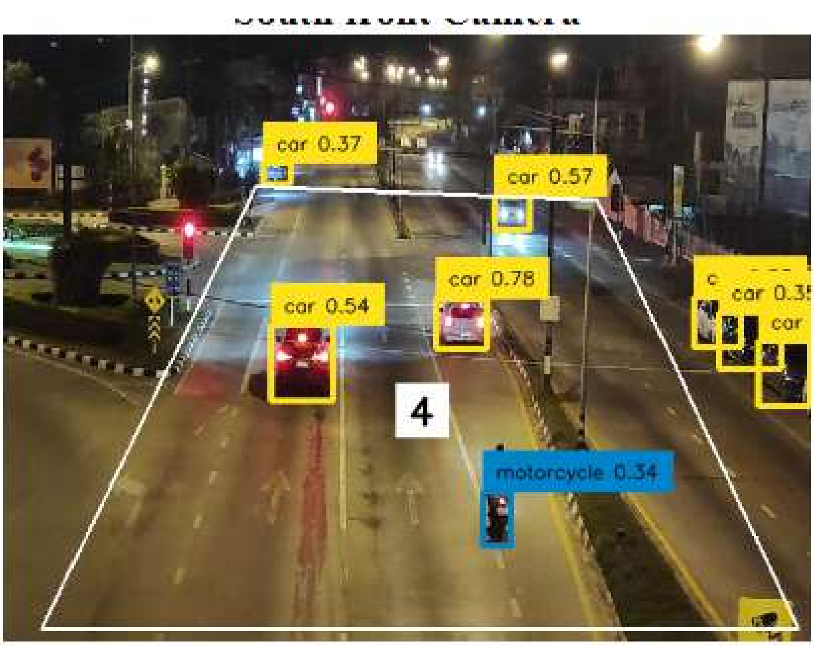


Figure 3.2.2 Analysis of the number of cars

From a camera with a car detection system, it is designed to measure the car behind the line to determine the time of the red light. If the number of cars is more than the line to be measured, then there is a pink box system to accommodate the excessive number of cars in the car. In the case of traffic jams, for example, if there is a car parked still in the box, it will be seen as a traffic jam and will increase the time to gradually pass more cars until the time limit is reached that no more cars can be reached. more than this.



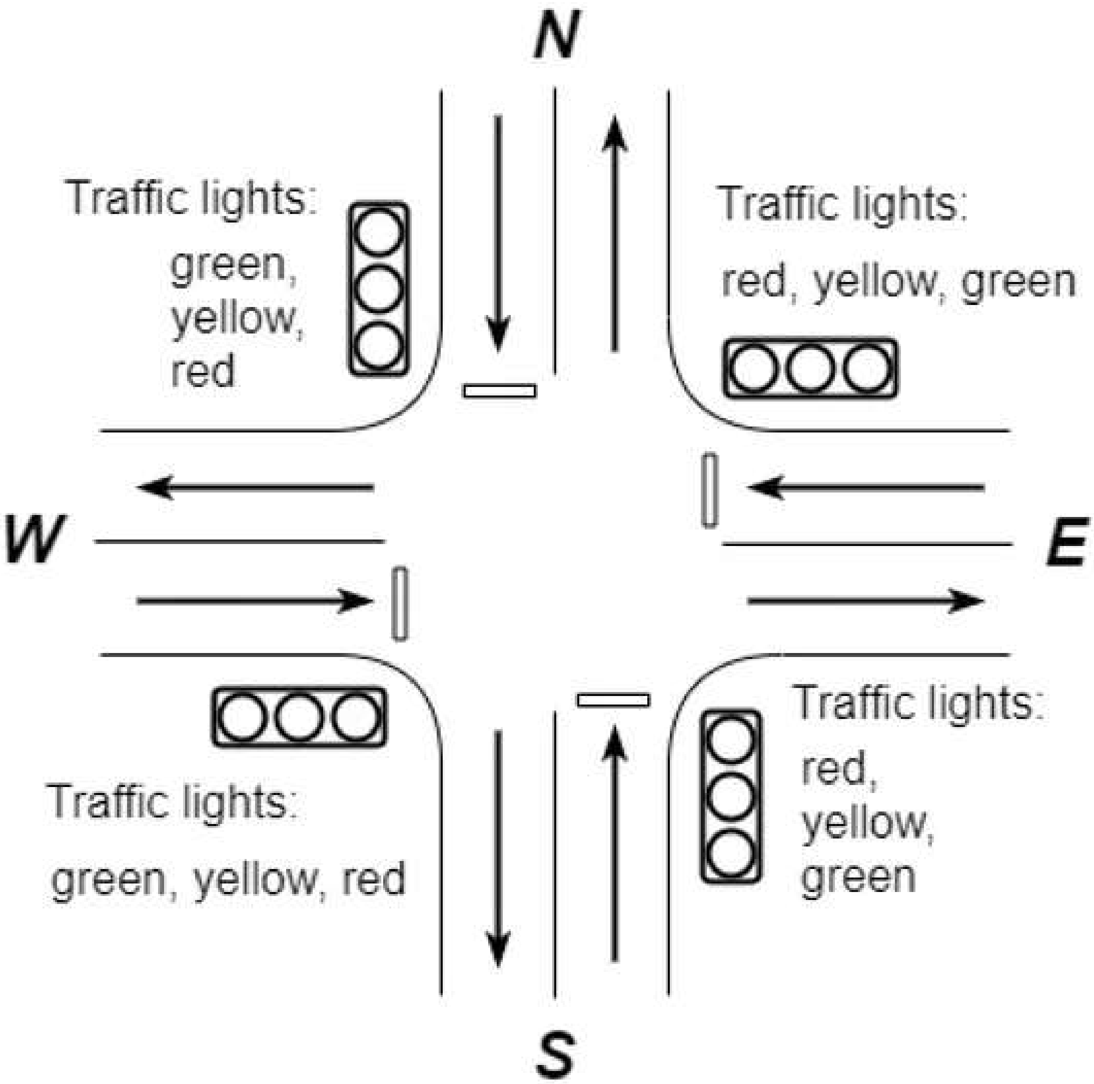


Figure 3.2.3(In the picture is a loop for traffic lights in the regular function.)

Example case

For example, in traffic number 1 the intersection in simulation.

- While side of number 2 is releasing the car will think of the time on the red arrow side to prepare for releasing the car in the next step
- On the side of number 3, there are 20 cars stuck, resulting in a red light that is right at 1.5 x 20 cars to get 30 seconds and add 5 seconds to give time to respond. More red lights for those caught in red lights. So there will be a time 35 seconds for the green light.
- And side number 4 will have more time to count the number of traffic jams than the red one and wait to think about when to release the car after intersection number 3. If at this time there are 10 cars at this side, it will be taken to calculate the time. Release the car following the number 3 site, it will take 20 seconds this way..
- In the event that there is a traffic jam on one side up to step 3.3 the number of cars in the pink frame will be added to reduce the number of cars stuck.

3.3 Use case diagram

After gathering user requirements, the new system could be represented by a use case diagram as shown in Figure 3.3

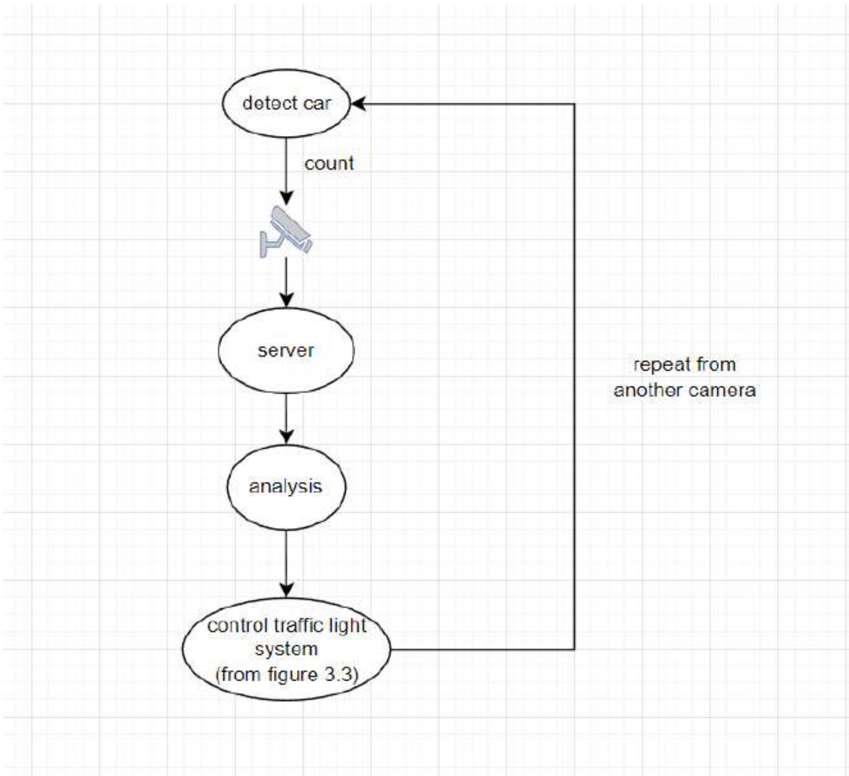


Figure 3.3 Use case diagram of the new system

3.4 Activity diagram

Analysis of a new system derives an activity diagram as shown in Figure 3.4

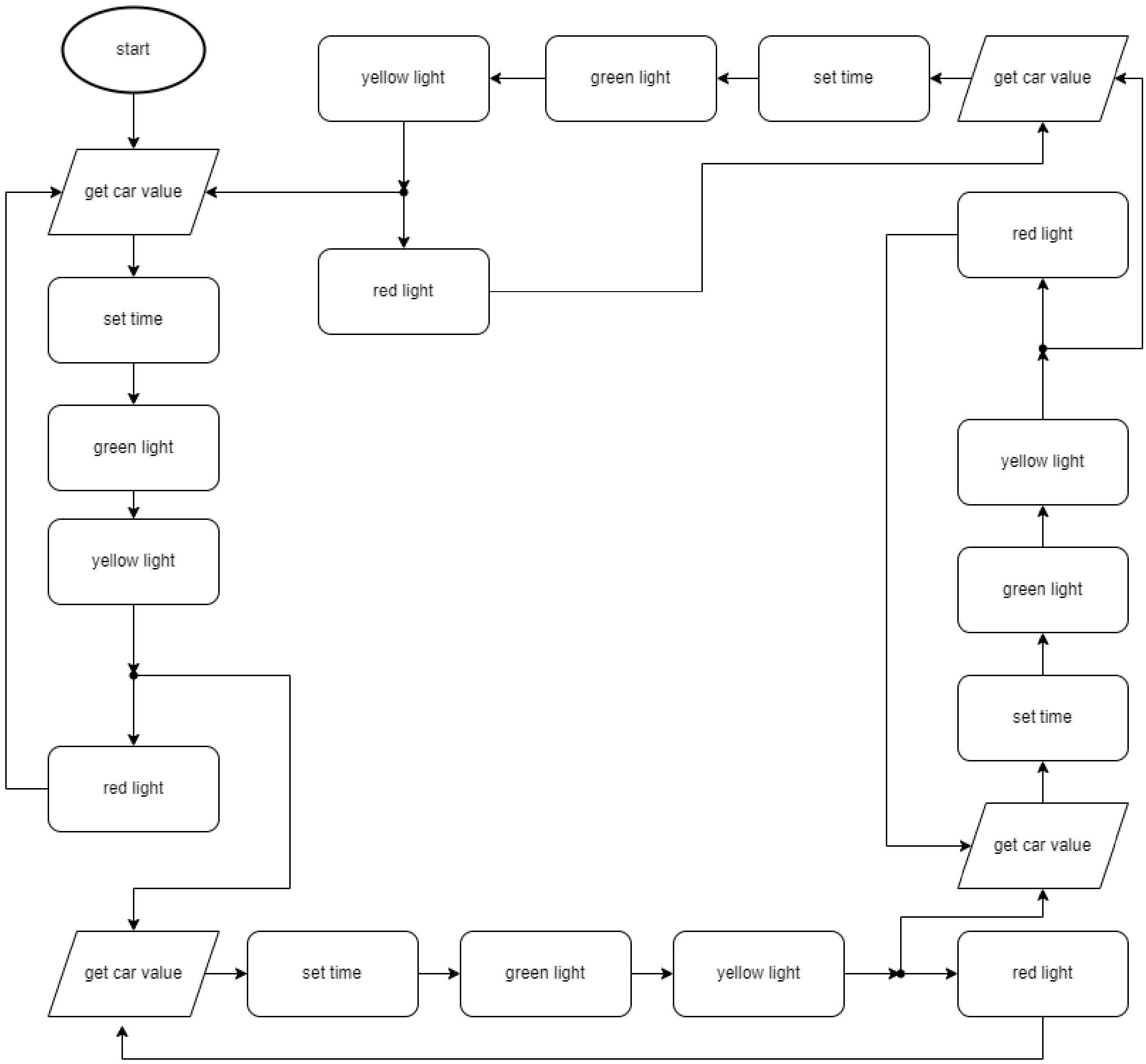


Figure 3.4 Activity diagram of the new system

3.5 Algorithm

In calculating the green light time of a traffic intersection, It uses vehicle detection by the YOLO V8 software to count the number of cars in calculating time so that time in traffic intersections is more flexible. By starting to take the number of vehicles into the calculation for green light time. The number of vehicles taken while the green light time of the previous traffic intersection is zero. Then the number of vehicles is used to calculate the time at the next traffic intersection. It will go on like this all the time. In the case of any traffic intersection, there are no vehicles that can be detected. It will cross that intersection and work on the next one to reduce traffic wait times and increase mobility. whereby the time at each traffic intersection has a minimum time limit in case a vehicle is detected. and the maximum time that the green traffic light can be displayed to control the time at each traffic intersection to be balanced. Using the formula for calculating time from below.

$$GT = \frac{(NoOfVehicle * VehicleTimeAverage)+6}{NoOfLanes}$$

GT is Green Signal Time

NoOfVehicle is number of vehicle from YOLO V8 to counting vehicle

VehicleTimeAverage is time average of vehicle to move

NoOfLanes is number of lanes intersection

6 is value of delay for the vehicle to set off

CHAPTER 4

EXPERIMENT AND RESULTS

4.1 Experiment

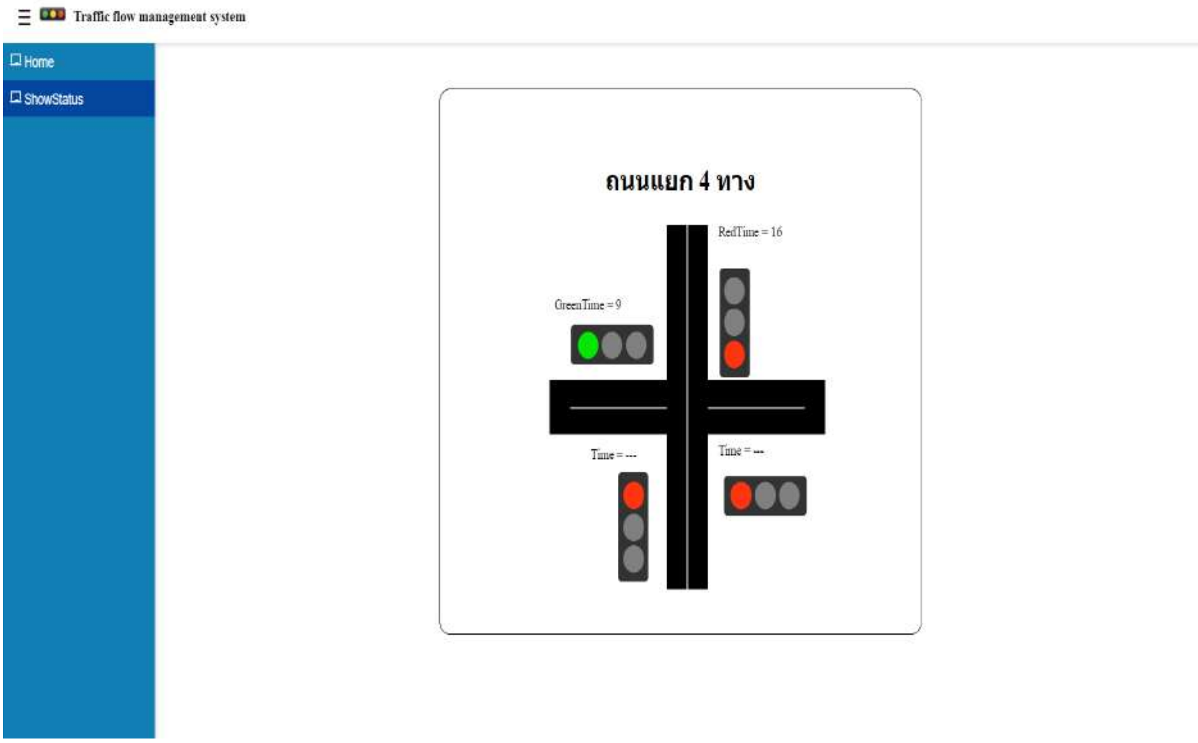
From the beginning, we wanted to make it real. But this cannot be done because creating a traffic light system requires a course in traffic engineering design to be able to change it and also requires using a real location. If an error occurs, it may cause widespread impacts. So we made a model. It is a concept experiment using a hypothetical scenario. To increase the efficiency of traffic movement and reduce congestion

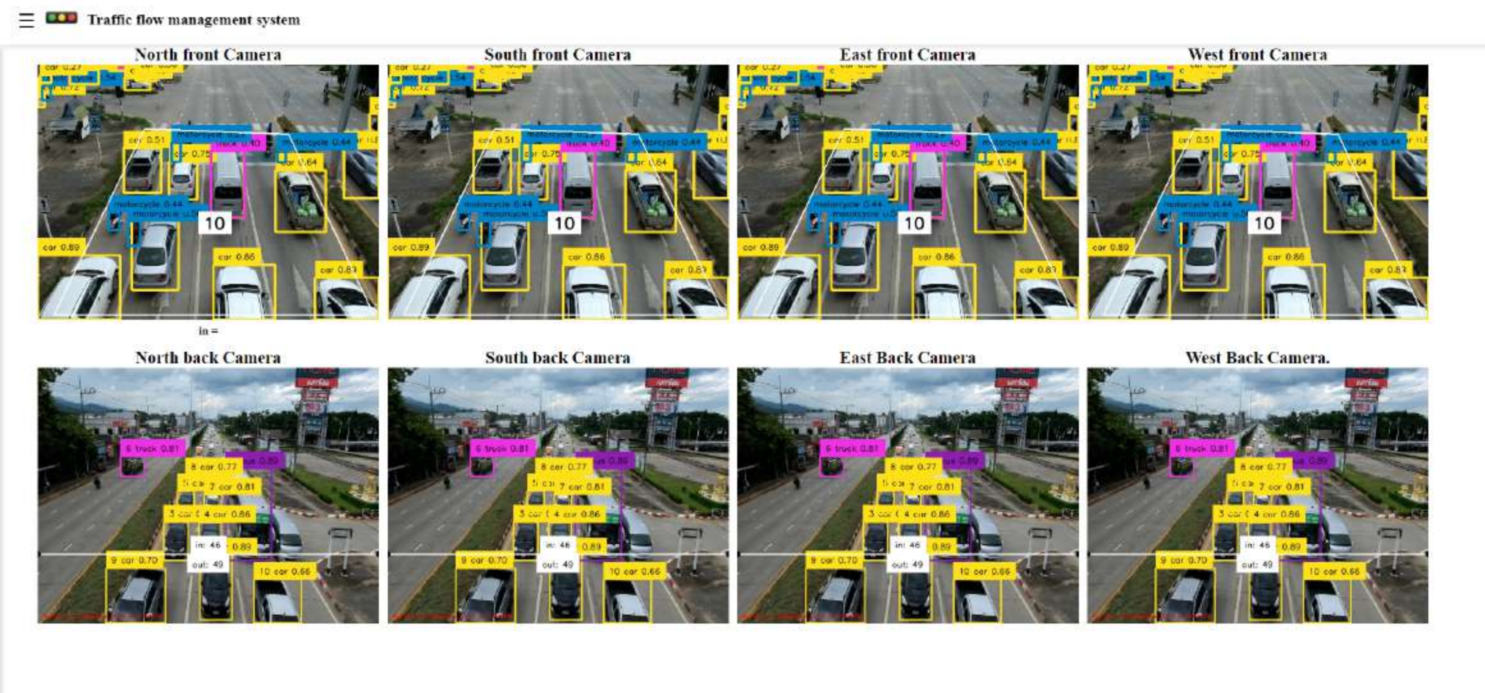
4.2 Results

Smart Intersection Traffic Lights systems are innovative solutions designed to improve traffic flow, and reduce congestion in urban areas. These systems use advanced technology to monitor traffic patterns, detect vehicles and help vehicles to flow easier by counting to calculate time per cycle.

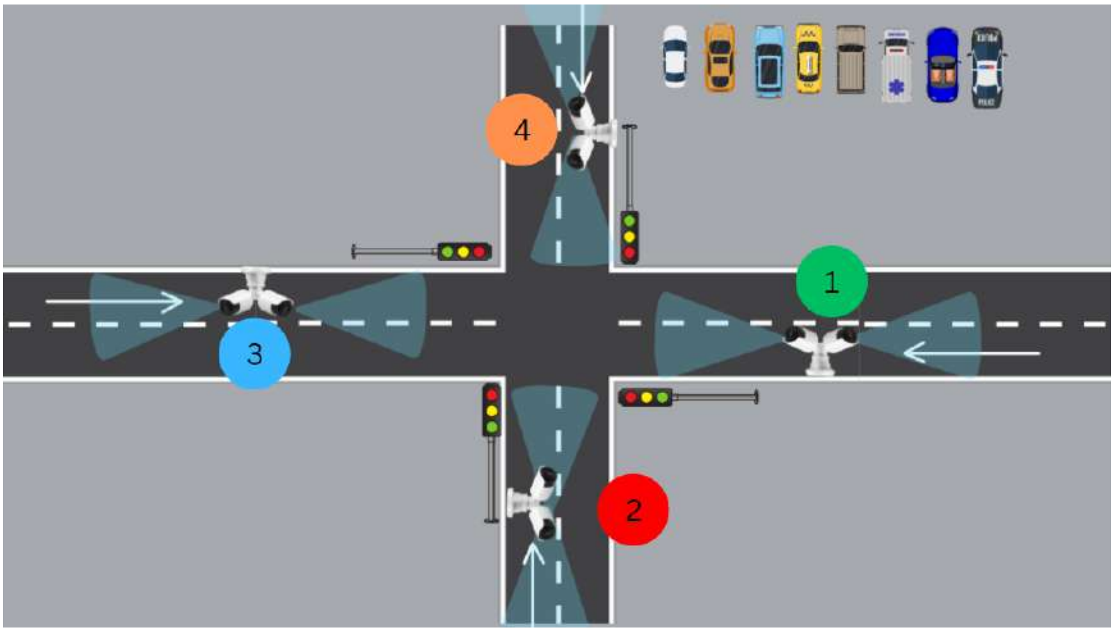
```
--------------------------------------------------------------
  RED TS 1 -> r: 141  y: 5  g: 3
GREEN TS 2 -> r: 0  y: 5  g: 1
  RED TS 3 -> r: 6  y: 5  g: 3
  RED TS 4 -> r: 133  y: 5  g: 3
--------------------------------------------------------------
  RED TS 1 -> r: 141  y: 5  g: 3Green Time:
 GREEN TS 2 -> r: 0  y: 5  g: 015

  RED TS 3 -> r: 5  y: 5  g: 3
  RED TS 4 -> r: 132  y: 5  g: 3
--------------------------------------------------------------
Green Time:  15
  RED TS 1 -> r: 140  y: 5  g: 3
GREEN TS 2 -> r: 0  y: 5  g: 0
  RED TS 3 -> r: 5  y: 5  g: 15
  RED TS 4 -> r: 132  y: 5  g: 3
--------------------------------------------------------------
  RED TS 1 -> r: 140  y: 5  g: 3
GREEN TS 2 -> r: 0  y: 4  g: 0
  RED TS 3 -> r: 4  y: 5  g: 15
  RED TS 4 -> r: 131  y: 5  g: 3
--------------------------------------------------------------
  RED TS 1 -> r: 139  y: 5  g: 3
GREEN TS 2 -> r: 0  y: 4  g: 0
  RED TS 3 -> r: 4  y: 5  g: 15
  RED TS 4 -> r: 131  y: 5  g: 3
--------------------------------------------------------------
```


Traffic flow management system
Home
ShowStatus
ถนนแยก 4 ทาง
RedTime = 16
GreenTime = 9
Time = ---
Time = ---

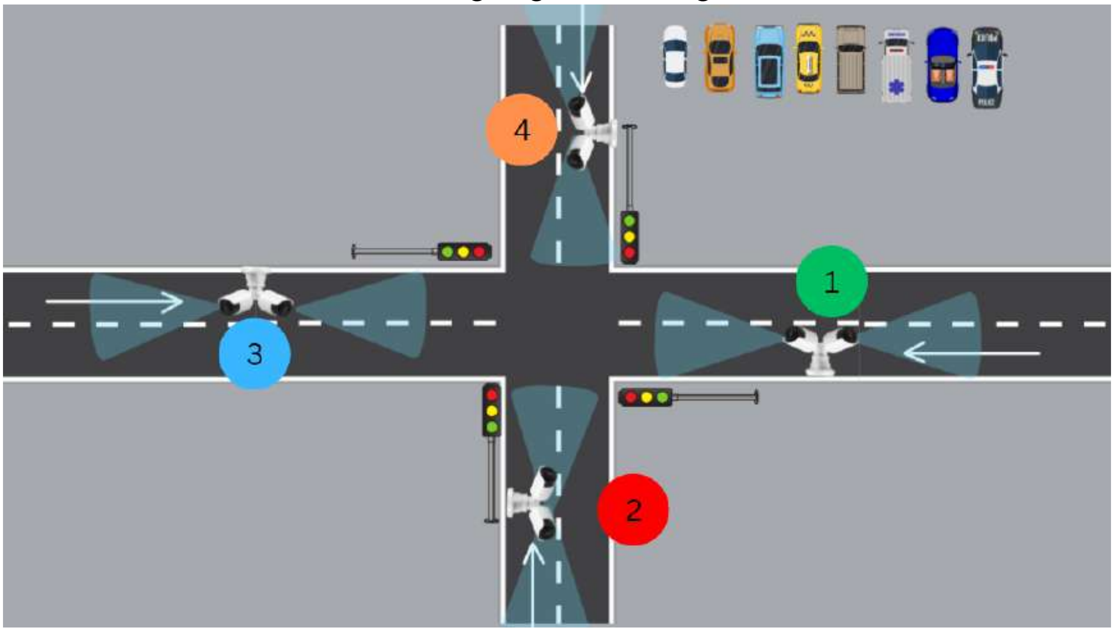

Traffic flow management system
North front Camera
South front Camera
East front Camera
West front Camera
in =
North back Camera
South back Camera
East Back Camera
West Back Camera.

Scenario 1 Checking the number of cars in the next traffic signal to calculate the time



From number 1, the light is green and is about to change to yellow. And when it changes to yellow until 0 seconds, it will check whether there is a car in number 3 or not. If not, it will go to check whether number 4 has a car or not, and so on. If there are no cars in any lane, it will change into the original loop.

Scenario 2 calculate the time to change signal traffic light



For example, from number 1 starting to change the traffic signal to green, it will count the number of cars and store the value until the traffic signal number 4 turns yellow at 0 seconds and the number of cars that have been stored will be used to calculate the time. of the traffic signal of number 1

Experiment in simulation.

To make the comparison of the fixed time traffic light and the adjustable time traffic light we need to find the difference between the number of the vehicles that crossed the intersection. so we created the simulation. The simulation shows the traffic light, time, vehicles, the number of vehicles that cross the intersaction and its direction.

the result : the old method

	round1	round2	round3	round4	round5	round6	round7	round8	round9	round10
lane1	142	147	145	149	155	158	157	138	134	129
lane2	95	84	81	85	82	84	91	95	105	107
lane3	23	29	23	32	31	26	18	24	26	31
lane4	30	34	38	26	28	23	26	33	31	32
total	290	294	287	292	296	291	292	290	296	299
vehicle/time	0.966667	0.98	0.956667	0.973333	0.986667	0.97	0.973333	0.966667	0.986667	0.996667

	round11	round12	round13	round14	round15	round16	round17	round18	round19	round20
lane1	134	145	133	139	129	125	143	133	153	125
lane2	92	92	89	107	100	91	81	107	94	100
lane3	30	24	27	26	30	27	31	19	26	31
lane4	38	30	35	29	38	38	31	36	27	42
total	294	291	284	301	297	281	286	295	300	298
vehicle/time	0.98	0.97	0.946667	1.003333	0.99	0.936667	0.953333	0.983333	1	0.993333

roundTotal
5854

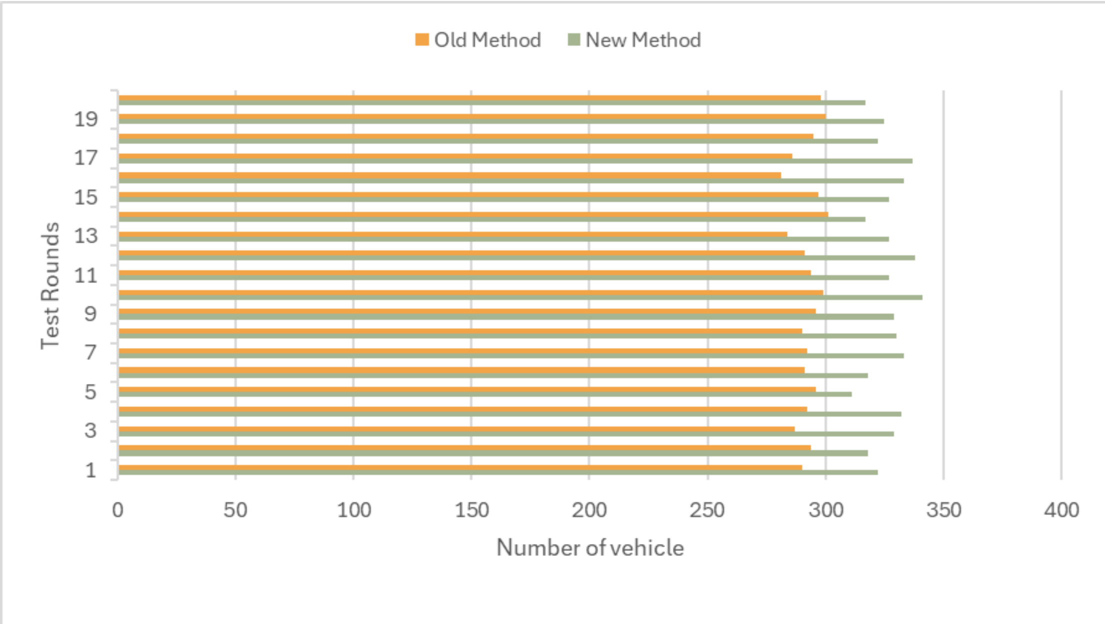
: the new method

	round1	round2	round3	round4	round5	round6	round7	round8	round9	round10
lane1	111	106	131	133	123	92	133	134	130	142
lane2	131	141	126	130	129	146	140	140	138	136
lane3	36	44	40	32	30	49	27	30	29	22
lane4	44	27	32	29	29	31	33	23	32	41
total	322	318	Chart Area	332	311	318	333	330	329	341
vehicle/time	1.073333	1.06	1.096667	1.106667	1.036667	1.06	1.11	1.1	1.096667	1.136667

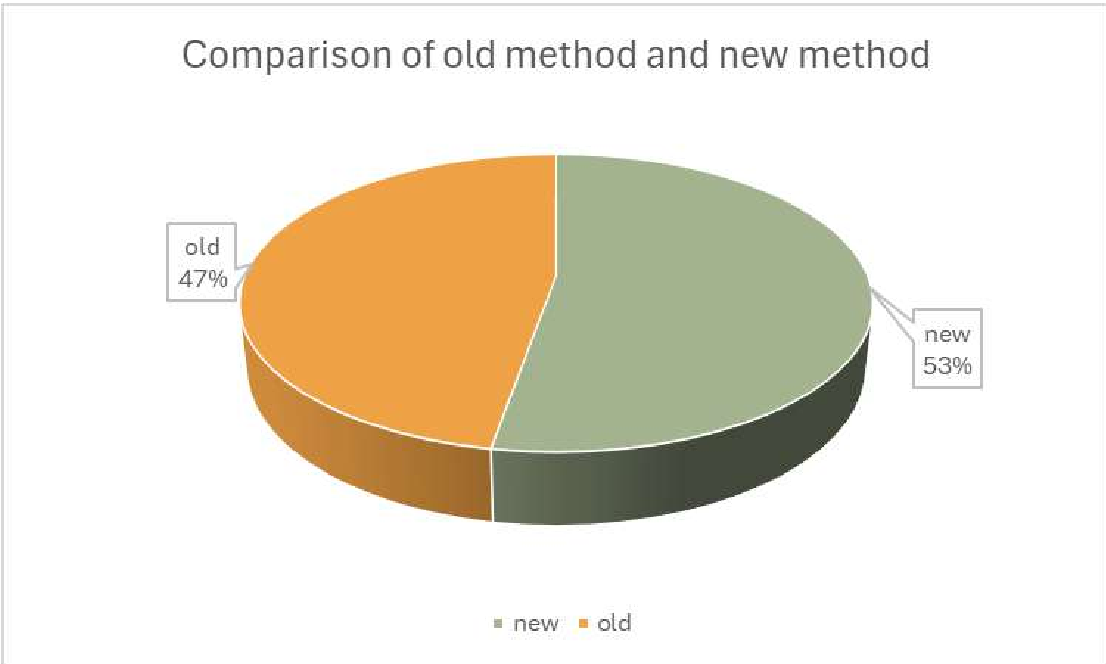
	round11	round12	round13	round14	round15	round16	round17	round18	round19	round20
lane1	112	134	126	111	141	134	134	135	123	118
lane2	132	142	146	137	129	136	140	121	139	133
lane3	37	19	31	37	20	31	44	34	33	30
lane4	46	39	24	32	37	32	19	32	30	36
total	327	338	327	317	327	333	337	322	325	317
vehicle/time	1.09	1.126667	1.09	1.056667	1.09	1.11	1.123333	1.073333	1.083333	1.056667

roundTotal
6533

the chart:



the comparison of the total number of vehicles that cross the intersection between the old method and the new method.



the comparison between old method and new method in percentage.

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

We have determined the scope of system development from planning the design of the traffic light system and found that. Relevant research to be used as a basis for analysis and design of the system for the field of this new traffic light system. In which we have tried to simulate the system where we have navigated the side of the software named YOLO V8 to help us. And what we have achieved is that we can count incoming and outgoing vehicles more accurately and it is more usable and efficient.

5.2 Discussion

Smart Intersection Traffic Lights systems are innovative solutions designed to improve traffic flow, increase pedestrian safety, and reduce congestion in urban areas. These systems use advanced technology to monitor traffic patterns, detect vehicles and pedestrians, and adjust traffic signals and crosswalk lights accordingly.

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APPENDIX

SOURCE CODE

```
"use client"
import React, { useEffect } from 'react';
interface Counter {
  in_count: number;
  out_count: number;

}
const Home: React.FC = () => {

  const host0 = "http://localhost:8000"
  const host1 = "http://192.168.124.234:8000"
  const host2 = "http://192.168.124.148:8000"

  return (
    <main style={{ "display": "flex", "flexDirection": "row",
"flexWrap": "wrap", "gap": "10px", 'textAlign': 'center', 'fontSize':
'20px', 'fontWeight': 'bold' }}>


      <div >
        <div id="cam1" >North front Camera</div>
        <div style={{ display: 'inline-block' }}>
          <img src={`${host0}/video_feed/1`} alt="" width={400} />
          <h6>in = {}</h6>
        </div>
      </div>
      <div>
        <div id="cam2" >South front Camera</div>
```

```
  <div style={{ display: 'inline-block' }}>
    <img src={`${host0}/video_feed/1`} alt="" width={400} />
  </div>
</div>
<div>
  <div id="cam3" >East front Camera</div>
  <div style={{ display: 'inline-block' }}>
    <img src={`${host0}/video_feed/1`} alt="" width={400} />
  </div>

</div>
<div>
  <div id="cam4" >West front Camera</div>
  <div style={{ display: 'inline-block' }}>
    <img src={`${host0}/video_feed/1`} alt="" width={400} />
  </div>

</div>
<div>
  <div id="cam5" >North back Camera</div>
  <div style={{ display: 'inline-block' }}>
  <img src={`${host0 }/video_feed/2`} alt="" width={400} />
  </div>

</div>
<div>
  <div id="cam6">South back Camera</div>
  <div style={{ display: 'inline-block' }}>
  <img src={`${host0}/video_feed/2`} alt="" width={400} />
  </div>

</div>
<div>
  <div id="cam7">East Back Camera</div>
  <div style={{ display: 'inline-block' }}>
```

```
      <img src={`${host0}/video_feed/2`} alt="" width={400} />
      </div>

    </div>
    <div>
      <div id="cam8">West Back Camera.</div>
      <div style={{ display: 'inline-block' }}>
      </div>
      <img src={`${host0}/video_feed/2`} alt="" width={400} />

    </div>
   </main>
  );
};
export default Home;
```